

CIS 434: Social Media and Text Analytics



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Tumblr at a Glance



PLATFORM

Large-scale microblogging platform for short-form content: text, images, GIFs, videos & links.



CONTENT SPREAD

Content diffusion driven by reblogging, tagging, and following mechanisms.



CORE CHALLENGE

Highly unstructured, text-heavy posts make moderation and analytics expensive at scale.



ANALYTICS GAP

Limited automated tools for creators to understand performance, audience, & engagement.

Target Audience

Our Main Target is to find Solution for Travel Content Creators on Tumblr



CREATORS

1. Travel vloggers
2. Travel bloggers
3. Tumblr influencers



NEEDS

1. Understand what drives engagement
2. Identify high-performing content
3. Improve reach & visibility



VALUE

1. Predict engagement performance
2. Identify winning topics & tone
3. Enable data-driven decisions

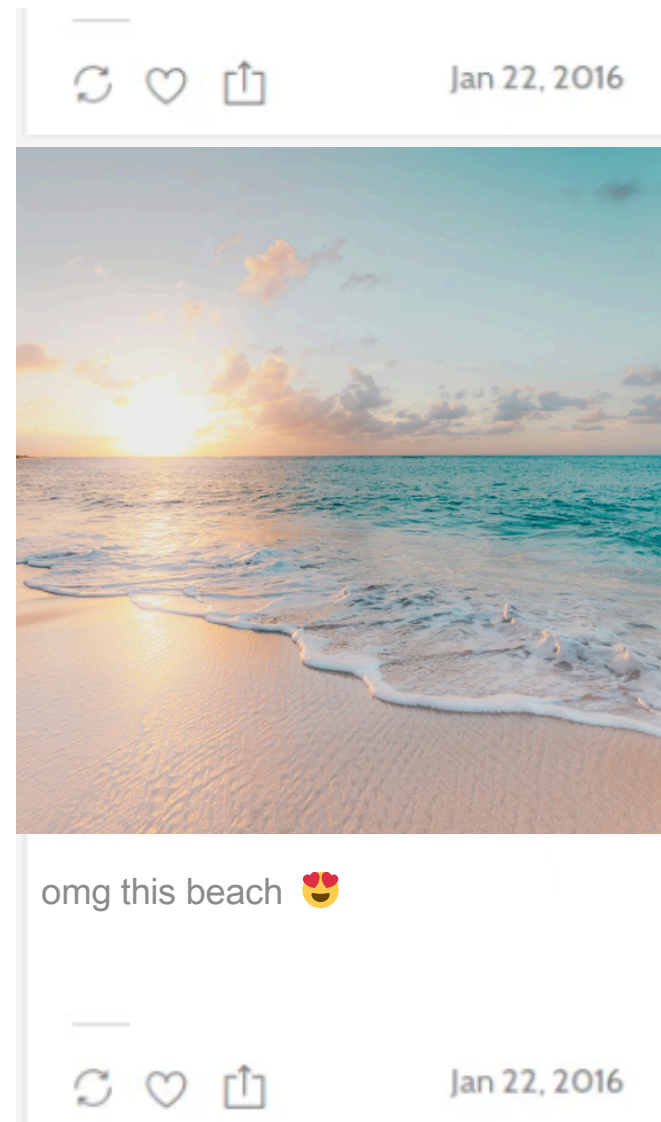
Problem Identification



- » Lack of structured analytics for creators
- » Engagement is unpredictable and difficult to evaluate
- » Titles and tags do not reflect engagement potential
- » Unstructured text limits meaningful analysis
- » Noisy, low-quality content distorts engagement signals

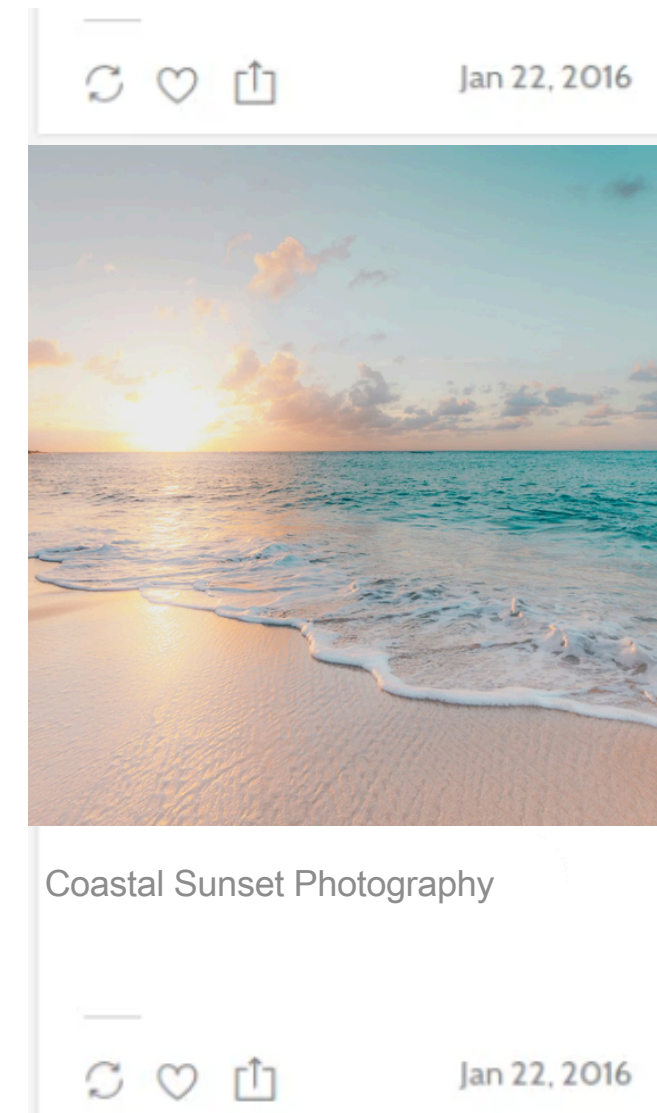
Tumblr creators cannot reliably predict high-engagement content due to a lack of structured analytics.

Vision



Current (Unstructured)

Sentiment: Neutral | Predicted: Low Engagement



Desired (Structured Insight)

Sentiment: Positive | Predicted: High Engagement

**What kind of sentiment and topics
lead to higher engagement on Tumblr
posts?**



Objective



Develop a model to predict high vs. low engagement based on sentiment, topic, and textual features.

Predict high vs. low engagement

Identify key drivers (sentiment & topics)

Enable data-driven content strategy

Data & Methodology

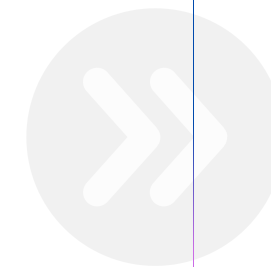
DATA COLLECTION



**PREPROCESSING
& CLEANING**



**FEATURE
ENGINEERING**
(SENTIMENT
+VADER + LDA +
TF-IDF)



**MODEL TRAINING
(XGBOOST)**

Cross-platform analysis using Tumblr (primary) and Reddit (validation dataset).

Data Overview

DATASET

Tumblr:
40,000+ posts
(primary dataset)
Reddit:
2,000+ posts
(validation
dataset)

01

02

ENGAGEMENT DEFINITION

Tumblr:
note_count

Reddit: upvotes +
comments

FEATURES USED

Sentiment(VADER)
Topic (LDA)
Text Length
TF-IDF (text features)

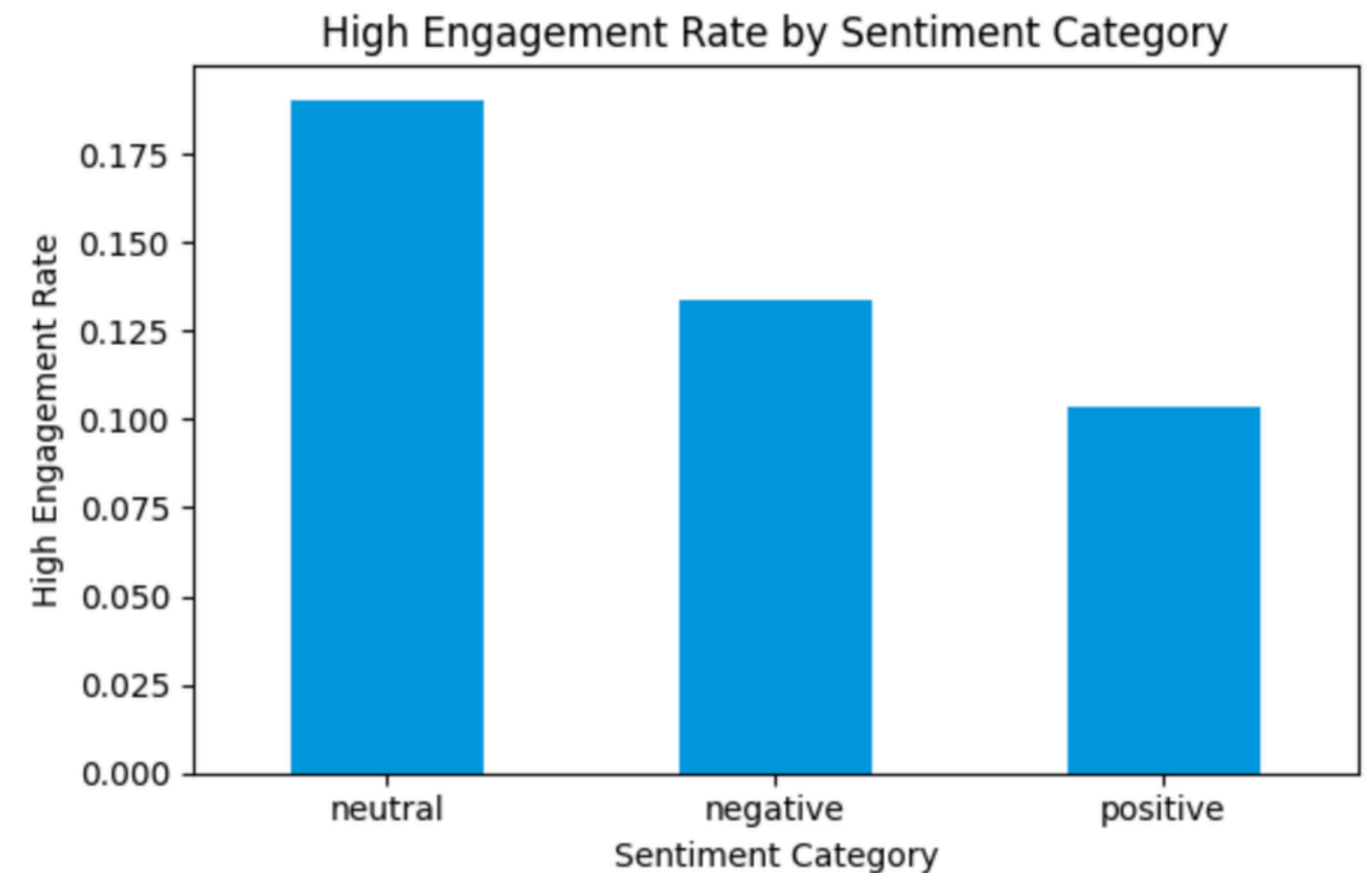
03



Sentiment vs Engagement

KEY INSIGHT

- Neutral posts have the highest engagement rate
- Positive posts show relatively lower engagement

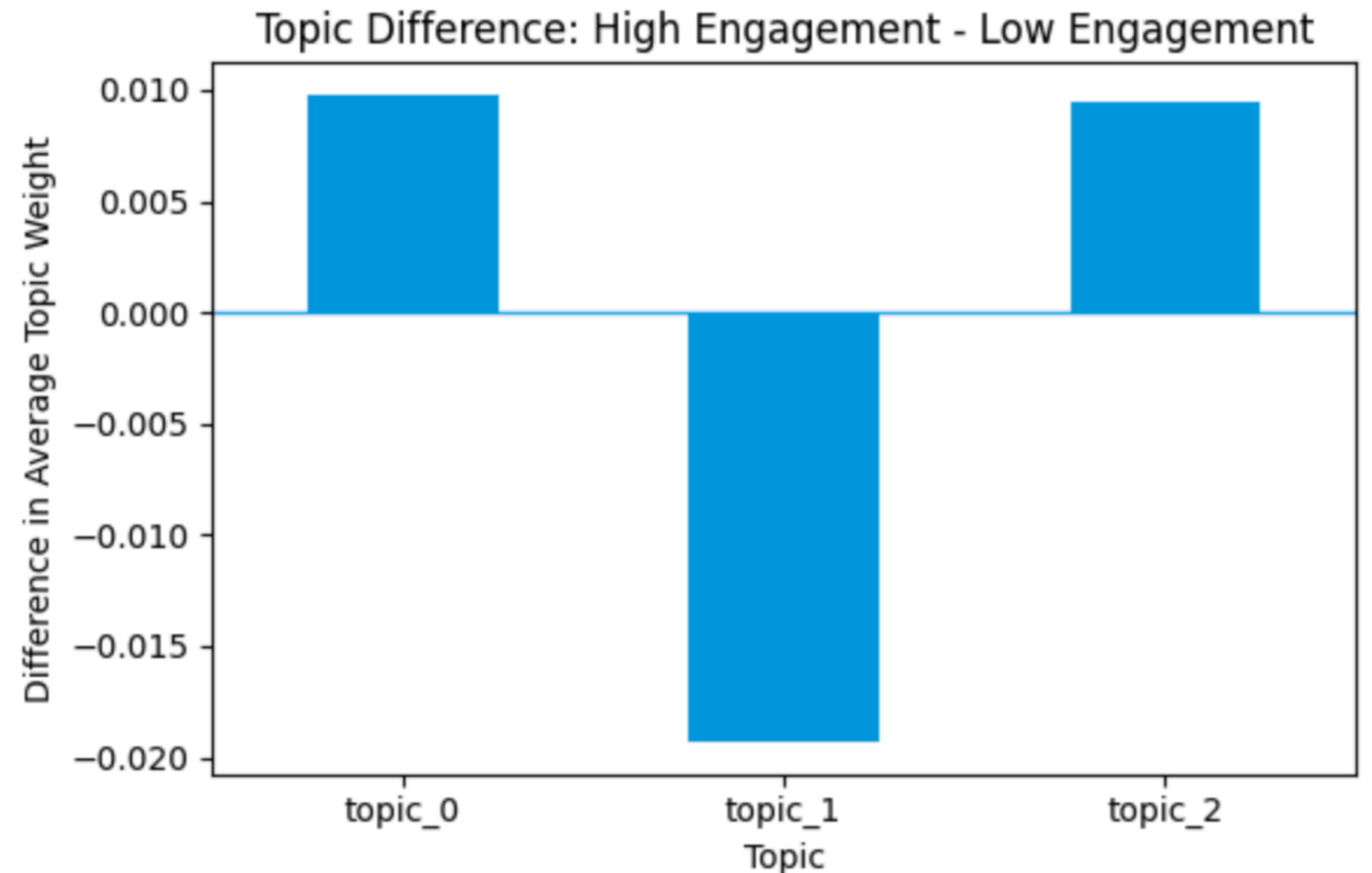


Tumblr engagement favors neutral, informative content over emotional content.

Impact on Engagement

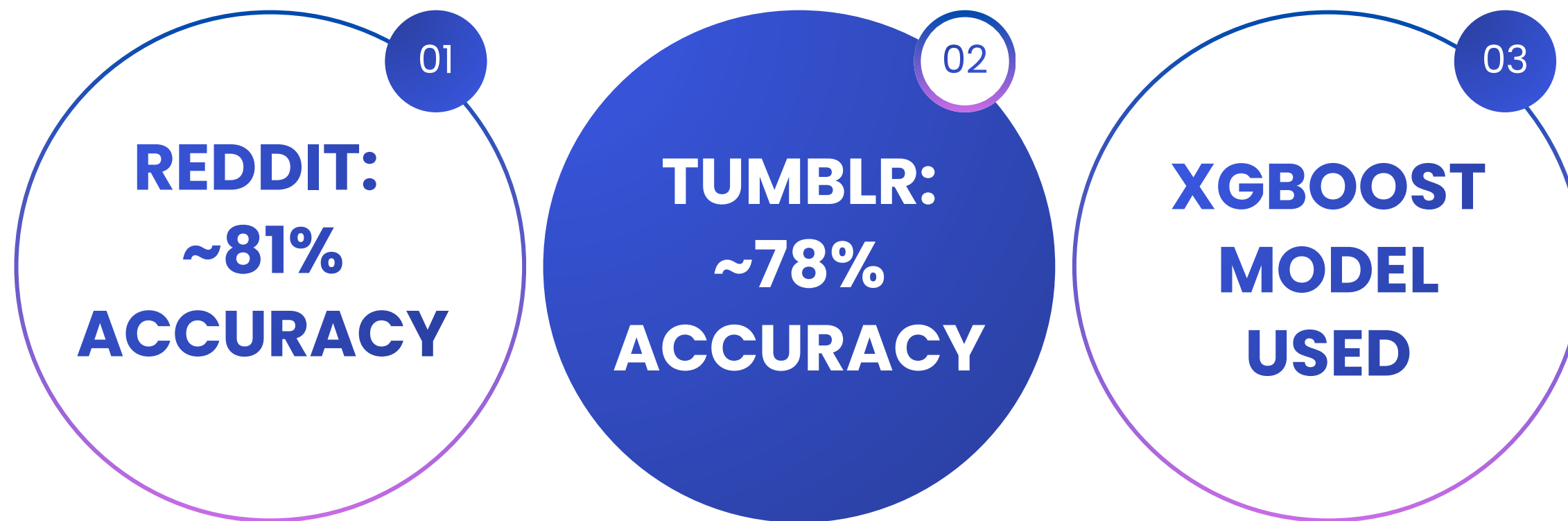
KEY INSIGHT

- Some topics show slight differences in engagement
- No topic consistently dominates performance



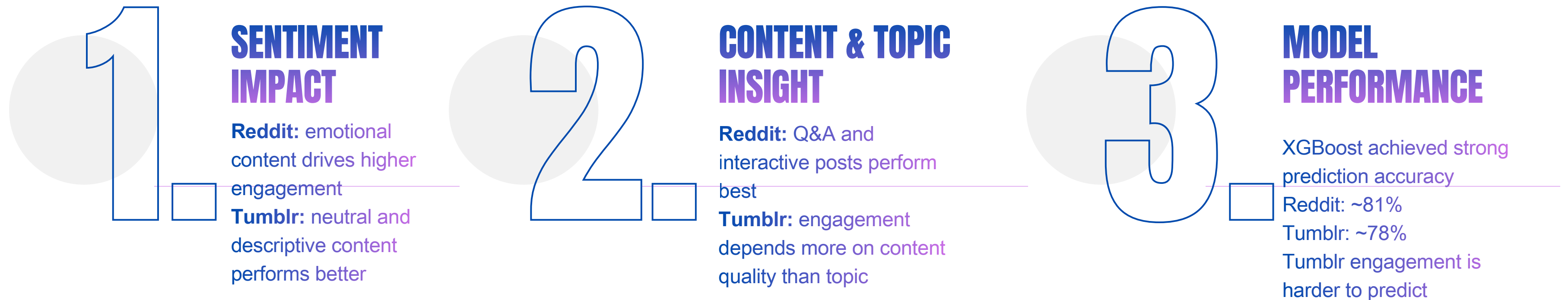
Content quality matters more than topic category on Tumblr.

Model Performance



Model performance shows that Tumblr engagement is more variable and harder to predict, confirming platform-specific differences.

Summarizing Our Key Results



Engagement drivers vary significantly across platforms.

Recommendation: Use Our App

From Insights to Action: Powered by Notecount.AI

PREDICT ENGAGEMENT BEFORE POSTING

- Real-time score
- Identify weak content **early**

UNDERSTAND WHY CONTENT UNDERPERFORMS

- Sentiment, topic, and structure **breakdown**
- Model explains key drivers

OPTIMIZE CONTENT INSTANTLY

- AI suggestions to improve **engagement**
- Rewrite captions for better **performance**

MOVE FROM GUESSWORK → DATA-DRIVEN POSTING



Limitations & Future Work



- » Focused only on travel-related content
- » Limited number of high-engagement posts
- » Simple features (TF-IDF, sentiment, topics)
- » May not generalize to other platforms or domains

Key Takeaway

Tumblr

Write descriptive,
neutral-toned content

Reddit

Lead with emotion &
interaction

Both

Use data before you
post

Notecount.AI turns this insight into actionable, predictive
decisions

Let's check out our App!

[Notecount.AI](#)

**Our model gives travel vloggers the
intelligence to post smarter on any
platform..**